

Vave Pipeline - Setup Guide

Overview

This guide explains how to set up and run the Vave Geospatial Risk Aggregation Pipeline.

Prerequisites

1. Unity Catalog main with appropriate permissions
2. Access to /Workspace/Repos/vave Git repository
3. Sample CSV files from sample_data folder in Git repo

Step 2: Upload Sample CSV Files

IMPORTANT: Files must be **manually uploaded** from the Git repository.

File Location in Git Repo

repo_root/
└── sample_data/

- └── vave_events_2026-04-20.csv
- └── vave_events_2026-05-01.csv
- └── vave_events_2026-05-05.csv

Upload Destination

Volume Path: /Volumes/main/_raw/external_data/vave_raw/

Upload Method

Databricks UI:

1. Navigate to **Data** → **Volumes**
2. Browse to main → _raw → external_data → vave_raw
3. Click **Upload**
4. Select all 3 CSV files from sample_data/ folder
5. Confirm upload

Expected CSV Format: event_id,event_timestamp,latitude,longitude,risk_score

EVT001,2026-04-20 08:15:23,51.5074,-0.1278,45.3

EVT002,2026-04-20 09:22:11,51.5155,-0.1426,62.8

Step 3: Verify File Upload

Action: Run this command to verify files are in the volume:

```
# In Databricks notebook
display(dbutils.fs.ls("/Volumes/main/_raw/external_data/vave_raw/"))
```

Expected output: 3 CSV files listed

Step 4: Run Bronze Ingestion

Notebook: Vave_Bronze_dbt

Purpose: Processes CSV files from volume into Bronze Delta table.

Action:

1. Open Vave_Bronze_dbt notebook
2. Set widget parameter: `fil_pth_str` = Comma-separated file paths

Example:

```
/Volumes/main/_raw/external_data/vave_raw/vave_events_2026-04-20.csv,
/Volumes/main/_raw/external_data/vave_raw/vave_events_2026-05-01.csv,
/Volumes/main/_raw/external_data/vave_raw/vave_events_2026-05-05.csv
```

1. Click **Run All**
2. Verify records inserted into `main.bronze.vave_eventsfeed`

Expected Output: === Bronze Ingestion - dbt Triggered ===



Target Table: `main.bronze.vave_eventsfeed`



File Type: csv



Files to Process: 3

✓ Inserted 190 records

✓ Processed 3 files

Note: The notebook tracks processed files in `main.control.processed_files` to prevent duplicate processing.

Step 5: Transform to Silver Layer

Purpose: Enrich Bronze data with H3 geospatial indexing.

Action: Run this SQL query in a notebook or SQL editor:

```
%sql
INSERT INTO main.silver.vave_eventsfeed
SELECT
    event_id,
    event_timestamp,
    latitude,
    longitude,
```

```

    risk_score,
    h3_latlng_to_cell_string(latitude, longitude, 8) AS h3_ix,
    DATE(event_timestamp) AS event_dt,
    source_file_name,
    ingestion_ts
FROM main.bronze.vave_eventsfeed
WHERE event_id NOT IN (SELECT event_id FROM main.silver.vave_eventsfeed);

```

What This Does:

- Adds H3 geospatial index at resolution 8 (~0.46 km hexagon edge)
- Extracts date from timestamp for partitioning
- Prevents duplicate records

Verify:

```

%sql
SELECT COUNT(*) FROM main.silver.vave_eventsfeed;
-- Expected: 190 records

```

```

SELECT DISTINCT h3_ix FROM main.silver.vave_eventsfeed;
-- Expected: ~57 unique H3 cells

```

Step 6: Refresh Gold Materialized Views

Purpose: Aggregate Silver data into analytics-ready Gold tables.

Action:

```

%sql
-- Refresh both gold materialized views
REFRESH MATERIALIZED VIEW main.gold.daily_risk_cell;
REFRESH MATERIALIZED VIEW main.gold.daily_risk_zone;

```

Verify:

```

%sql
SELECT * FROM main.gold.daily_risk_cell
ORDER BY event_dt DESC, total_risk_score DESC
LIMIT 10;

```

Expected Output:

- Daily risk aggregations by H3 cell
- Multiple dates from sample data (April 20 - May 9, 2026)
- 57 unique H3 cells in London area
- Total of 190 events

Step 7: View Dashboard

Dashboard: Vave Geospatial Risk Analysis NM

Widgets:

1. **Total Events** counter: 190
2. **Unique H3 Cells** counter: 57
3. **Average Risk Score** counter: 47.32
4. **Risk Trend Over Time**: Line chart showing daily total risk scores
5. **Top 10 Highest Risk Cells**: Bar chart of H3 cells with highest cumulative risk
6. **Top 20 Risk Records**: Table showing detailed records

Data Sources:

- main.gold.daily_risk_cell
- main.gold.daily_risk_zone

Access: Navigate to Dashboards in workspace or use published dashboard URL

Troubleshooting

Files Not Processing

Issue: Files uploaded but not appearing in Bronze table

Solution:

1. Verify files in volume:

```
dbutils.fs.ls("/Volumes/main/_raw/external_data/vave_raw/")
```

- 2 .Check widget parameter fil_pth_str is set correctly in Bronze notebook

1. Verify files not already in control table:

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H3 Function Not Found

Issue: h3_latlng_to_cell_string function error

Solution: Ensure Databricks Runtime 13.3 LTS or higher (H3 functions built-in)

Gold Views Empty

Issue: Gold materialized views show no data **Solution:**

1. Verify Silver table has data

```
%sql  
SELECT COUNT(*) FROM main.silver.vave_eventsfeed;
```

1. Refresh materialized views manually
2. Check H3 index column populated:

```
%sql
```

```
SELECT h3_ix FROM main.silver.vave_eventsfeed WHERE h3_ix IS NULL;
```

Dashboard Shows Old Data

Issue: Dashboard not reflecting recent data

Solution:

1. Refresh materialized views
2. Refresh dashboard page

Daily Production Workflow

Batch Schedule (Recommended: Daily at 2 AM):

1. **File Arrival:** CSV files land in volume throughout previous day
2. **Bronze Ingestion:** Scheduled job runs Vave_Bronze_dbt notebook
 - Discovers new files
 - Processes only unprocessed files (idempotent)
 - Updates control table
3. **Silver Transformation:** Scheduled job enriches with H3 indexing
4. **Gold Refresh:** Materialized views refresh
5. **Dashboard Update:** Dashboard displays updated metrics

Orchestration Options:

- **Databricks Jobs** (recommended) - Native scheduling
- **dbt Cloud** - Data transformation orchestration
- **Apache Airflow** - Complex workflow management
- **Azure Data Factory** - Azure-native orchestration

No Auto-Trigger: Files landing in volume do NOT automatically trigger pipeline. Manual or scheduled execution required.

Migration to Vave Workspace

Target Structure: /Workspace/Repos/vave/notebooks/

Essential Files:

1. Vave_Config_Params.py - Shared configuration
2. Vave_Setup.py - Infrastructure setup
3. Vave_Bronze_Ingestion.py - Main ingestion notebook

Migration Steps:

1. Export all 3 notebooks as .py source files
2. Commit to Git repository at /Workspace/Repos/vave/notebooks/
3. Update %run paths to relative paths (e.g., ./Vave_Config_Params)
4. Run setup notebook in new workspace
5. Test with sample files

Dashboard Migration:

- Export as JSON from current workspace
- Import to Vave workspace
- Reconnect datasets to Unity Catalog tables